



CARPENTRY

HPC Carpentry

An Overview of HPC Carpentry's Activities

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A Very Brief History of The Carpentries

- ▶ Software Carpentry was founded in 1998 with the mission of teaching lab skills for research computing by Greg Wilson and Brent Gorda.



Dr Greg Wilson
<https://carleton.ca/scs/?p=14196>

Where does HPC Carpentry fit?

1. Knowing about version control and functions does not quite solve all problems.
2. Researchers with HPC requirements do not necessarily know how to effectively scale their applications
3. HPC facility operators become frustrated by non-knowledgable users making use of shared resources.
4. Solution: Use Carpentries techniques.

How did it all begin

- ▶ Earliest commit to hpc-carpentry GitHub 2013 by Compute Canada
- ▶ Peter Steinbach blog post (2017) - HPC in a day
- ▶ CarpentryCon 2018, 2020, 2022
- ▶ Super Computing BoF 17, 18, 19, 21
- ▶ Winter 2021, accepted into Carpentries Incubator
- ▶ Begin working towards Lesson Program Incubation in 2021 and formally accepted to the Lesson Program Incubation in June 2023
- ▶ April 2024, workflow lessons enters testing

Where are we now?

1. We are doing things In the Carpentries Way
 - ▶ two day setting
 - ▶ students type along with instructors
 - ▶ get feedback
2. Two main takeaways:
 - ▶ muscle memory
 - ▶ vocabulary
3. Not experts after two days but:
 - ▶ know what answer looks like
 - ▶ know how to find answers



- ▶ Core curriculum: Using High-Performance Computing Systems, HPC Workflow Management with Snakemake
- ▶ Recommend to start with the SWC Unix Shell lesson where learners are not already familiar with working in the shell
- ▶ An extended curriculum of other lessons

Current Status: Lesson Support

- ▶ Instructor Onboarding documentation available
- ▶ Lesson Maintainers onboarded by The Carpentries
- ▶ Infrastructure support allows customisation of lessons to reflect local cluster configurations



Current Status: Community

- ▶ Welcomed many new members to regular community calls + Maintainer teams
- ▶ Gathered interest in teaching+hosting workshops from the wider Carpentries community
- ▶ Completed Steering Committee elections

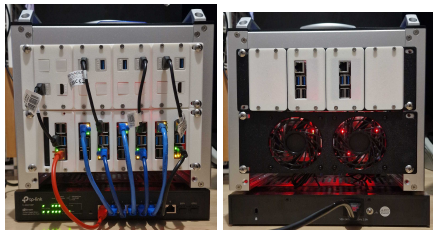


Current Status: Lesson Program Adoption

- ▶ Adoption as an official lesson program of The Carpentries was planned for 1 September 2026
- ▶ Changes in Carpentries Core Team support called for a short delay
- ▶ Adoption will be completed between now and November 2026

A miniHPC - CarpentriesOffline

- ▶ Pixie: Raspberry Pi 4s, one login node, five compute nodes, one access point
- ▶ The same open source software stack as many HPCs: Slurm, lmod, munge
- ▶ Running Workshops and Hackerthons
 - ▶ Lesson to build a login node and a compute node
 - ▶ Adapt HPC lesson for Pixie
 - ▶ Networking lesson



Front and Back



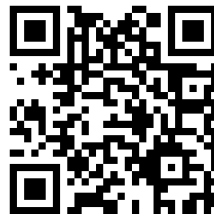
CARPENTRIES

Comms Channels

- ▶ Come talk to me
- ▶ Visit our website
- ▶ Join our bi-weekly meetings
- ▶ Join our channel in The Carpentries' Slack workspace



HPC Carpentry



CarpentriesOffline



Your Help is Needed

- ▶ Do you have any suggestions for more domain-specific conferences where we could reach members of HPC Carpentry's target audience of learners and or potential instructors hosts for workshops?
- ▶ Are you interested in teaching, organising and/or hosting HPC Carpentry workshops?
- ▶ Are you interested in getting involved with maintaining the lessons?



- ▶ **Greg Wilson's talk at PyCon 2014**
<https://www.youtube.com/watch?v=Wv0rf3Fyh7M>
- ▶ **Andrew Reid's talk at SIGHPC 2024**
<https://www.youtube.com/watch?v=FtKO619O5g0>
- ▶ **Peter Steinbach's blog post, HPC in a day?**
<https://carpentries.org/blog/2017/06/hpccarpentry>
- ▶ **HPC Carpentry Website** <https://www.hpc-carpentry.org>
- ▶ **CarpentriesOffline for the miniHPC** <https://carpentriesoffline.org>
- ▶ ChatGPT for its help with the $\text{\LaTeX}$ template!